

Sparse Connective and Event Driven CNN Accelerator for Real-Time Object Detection

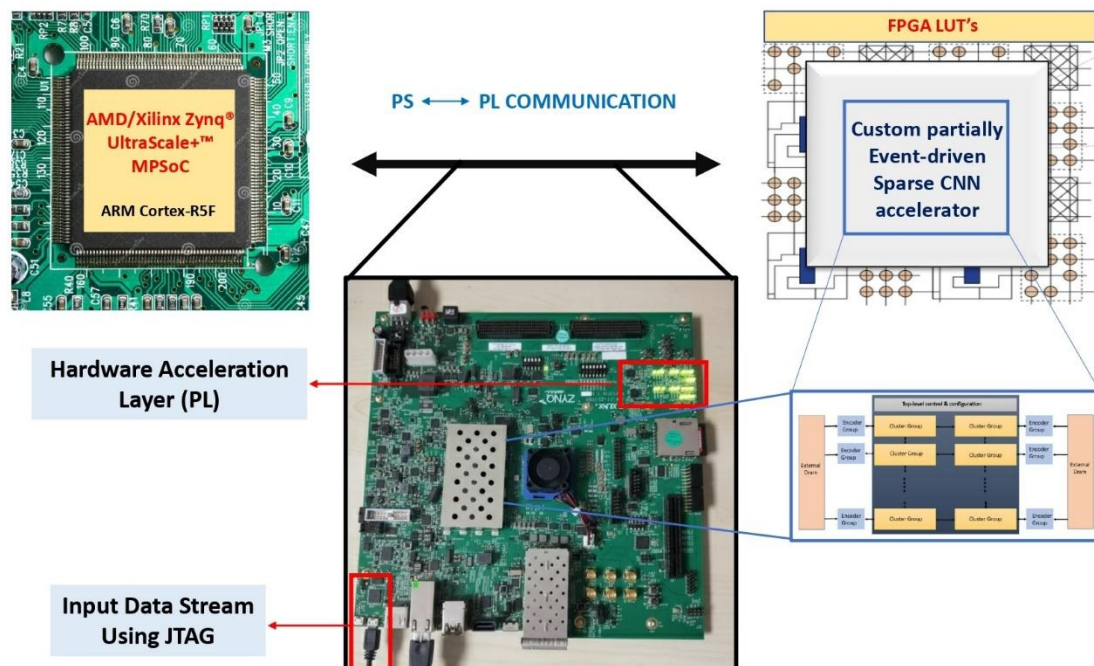
1. Introduction

Real-time object detection on edge devices is traditionally constrained by high computational load, power consumption, and latency due to frame-based CPU inference pipelines. Conventional systems process complete image frames continuously, even when significant visual changes are absent, resulting in inefficient resource utilization.

This project presents a **low-latency, energy-efficient event-driven hardware accelerator** for real-time object detection. Unlike traditional frame-based execution, the proposed architecture activates computation only when meaningful events occur, significantly reducing redundant operations, memory access, and switching activity.

The accelerator is validated on FPGA and designed with scalability toward ASIC implementation.

Sparse Connective and Event Driven CNN Accelerator for Real-Time Object Detection



2. Project Objectives

- Design a **real-time object detection accelerator**
- Implement a fully **event-driven execution model**
- Reduce **latency below 2.5 ms**
- Minimize **power consumption (<5W)**
- Eliminate redundant frame-based computation
- Demonstrate FPGA validation on ZCU102

3. System Architecture Overview

The proposed architecture replaces frame-based processing with an event-driven pipeline.

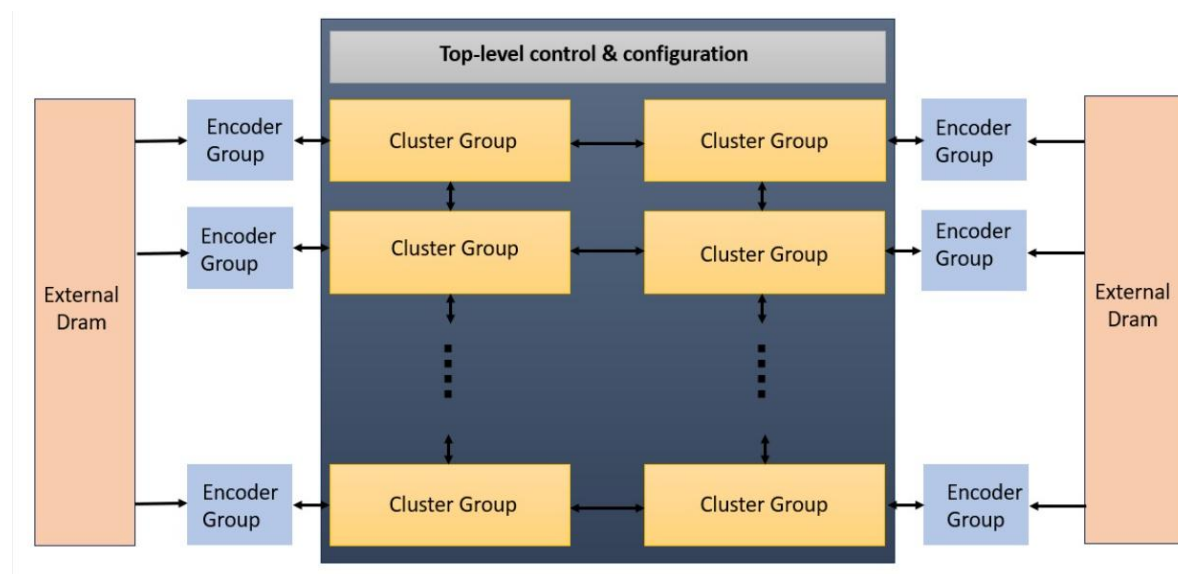
3.1 Event-Driven Processing Model

Computation is triggered only by:

- Pixel intensity changes
- Motion-based Regions of Interest (Rols)
- Feature activation thresholds

This eliminates unnecessary processing of static frame regions.

Top Level Architecture



3.2 Accelerator Pipeline

The proposed accelerator follows a fully event-driven, streaming pipeline architecture, where each stage operates asynchronously and is activated only upon the arrival of valid events. Unlike conventional frame-based systems that require complete frame buffering and global synchronization, this pipeline eliminates full-frame dependency and processes sparse event streams in real time.

The complete accelerator consists of four major stages:

1. Event Generator

The Event Generator acts as the front-end interface between the image sensor (or host PC) and the hardware accelerator. Its primary role is to convert dense pixel data into compact, sparse event representations.

Key Functions:

- Detects significant pixel-level changes or motion regions.
- Applies configurable intensity or gradient thresholds.
- Encodes events in the format:
(x-coordinate, y-coordinate, timestamp, polarity/value).
- Suppresses static background regions to eliminate redundant computation.
- Buffers events temporarily before forwarding to the scheduler.

Instead of forwarding full image frames, only meaningful pixel activations are converted into events. This dramatically reduces input bandwidth and memory requirements.

2. Event Scheduler

The Event Scheduler manages asynchronous event traffic and dynamically distributes incoming events to available Processing Elements (PEs).

Key Responsibilities:

- Maintains a global event queue.
- Handles bursty and irregular event arrival patterns.
- Implements load-balancing across PE clusters.
- Prevents pipeline stalls using handshake-based flow control.
- Prioritizes events when multiple regions are active.

The scheduler ensures that hardware resources are efficiently utilized while preserving event ordering and layer-level dependencies.

3. Processing Element (PE) Array

The Processing Element Array forms the computational core of the accelerator. It consists of multiple event-aware PEs organized in a scalable cluster architecture.

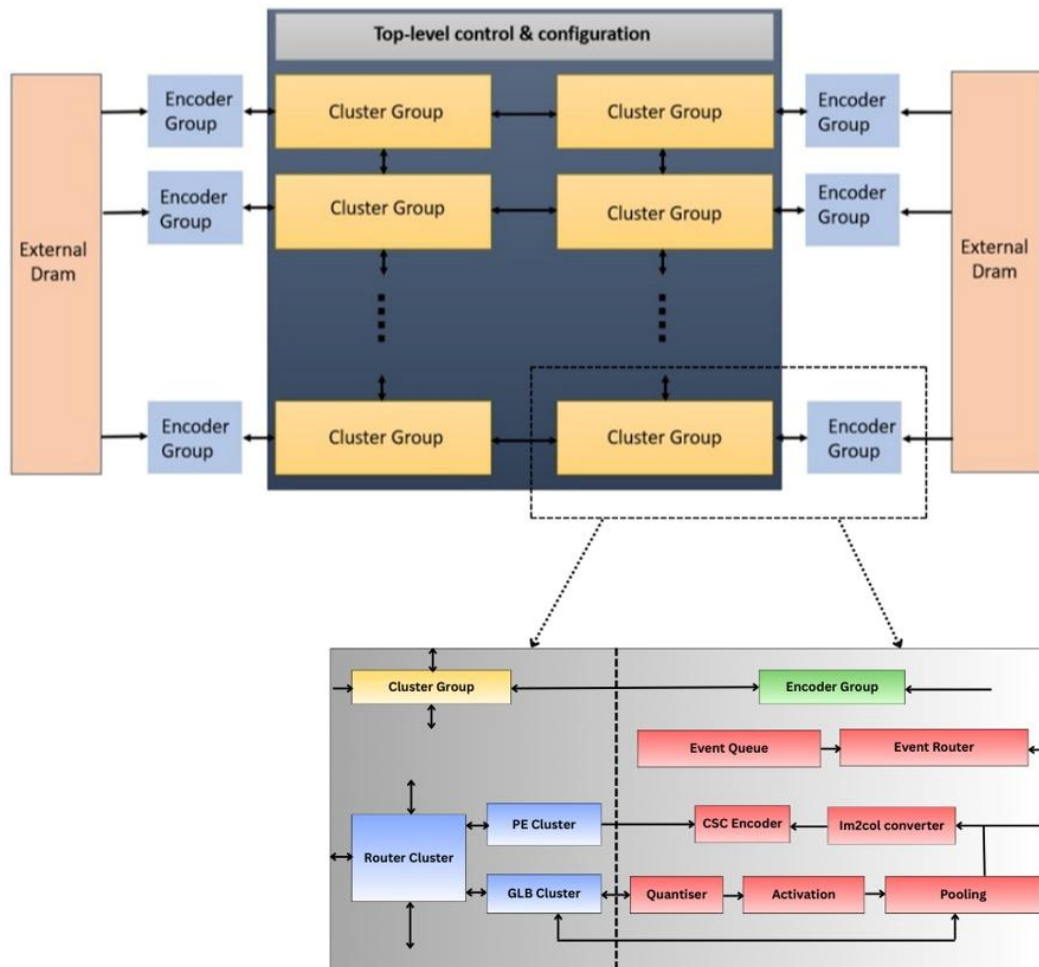
Key Characteristics:

- Each PE remains idle until a valid event is received.
- Performs event-triggered convolution and feature extraction.
- Accumulates sparse partial sums (psums).
- Supports weight-stationary dataflow for improved locality.
- Operates in parallel for multiple independent events.

Since computation is triggered only when events are valid, idle switching activity is minimized, significantly reducing dynamic power consumption.

The PE array supports:

- Convolution operations
- Activation functions (e.g., ReLU-like simplified logic)
- Sparse accumulation
- Intermediate feature propagation



4. Detection Output Module

The Detection Output Module aggregates partial results from the PE array and generates the final detection outputs.

Core Functions:

- Collects psums from multiple clusters.
- Applies thresholding and classification logic.
 - Generates:
 - Object class label
 - Bounding box coordinates
 - Confidence score
- Formats output for host transmission.

Since outputs are generated in a streaming manner, detection results can be produced immediately after sufficient evidence is accumulated, without waiting for full-frame completion.

Asynchronous Streaming Dataflow

A major strength of this pipeline is its asynchronous event-driven execution:

- No global frame synchronization required.
- Each stage communicates using Valid and Ready handshake signals.
- Data flows continuously as events propagate.
- Latency depends on event density, not frame size.

This enables:

- Sub-millisecond internal processing latency.
- Immediate response to dynamic scene changes.
- Efficient handling of sparse environments.
- Reduced memory bandwidth usage.

Pipeline Advantages Over Frame-Based Systems

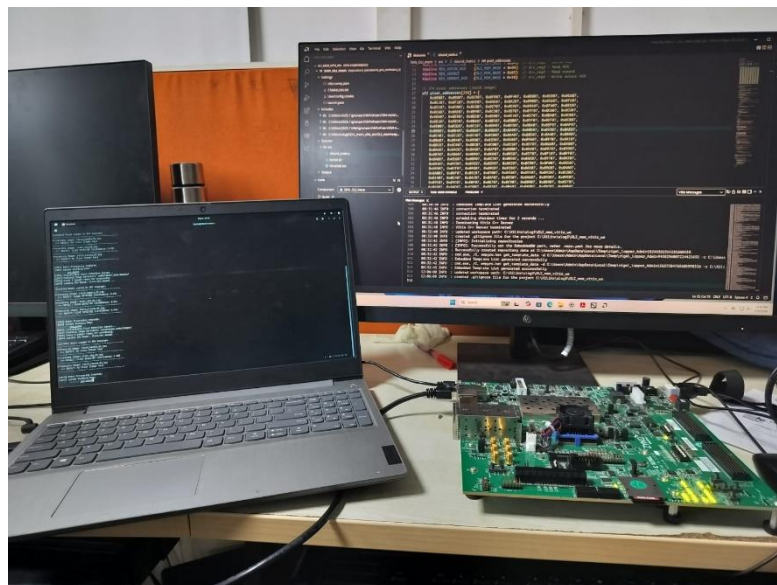
Feature	Frame-Based Pipeline	Event-Driven Pipeline
Input Processing	Full frame	Sparse events only
Synchronization	Global frame sync required	Fully asynchronous
Latency	Frame-dependent	Event-dependent
Computation	Continuous	On-demand
Power Usage	High	Significantly reduced

The accelerator pipeline therefore enables real-time, low-latency object detection, where computation scales with scene activity rather than image resolution. This makes the architecture particularly suitable for edge devices operating under strict power and timing constraints.

4. FPGA Implementation

The accelerator was implemented on:

- **Xilinx ZCU102 FPGA**
- Synthesized using **Xilinx Vivado**
- 2x2 Event Cluster configuration for validation



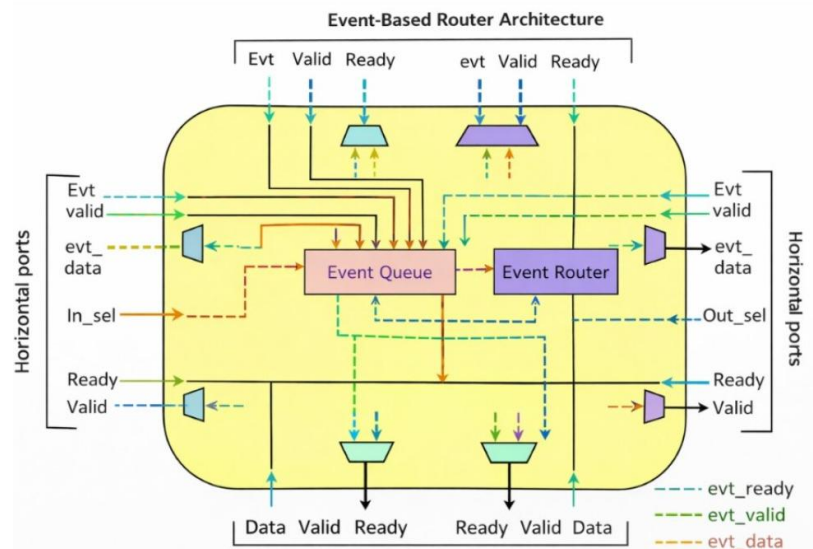
Verification Setup

- Host PC generates event streams
- UART-based asynchronous communication
- FPGA processes sparse events in real-time
- Detection results returned with ultra-low latency

This setup validates realistic deployment scenarios for edge vision systems.

5. Event-Based Router Architecture

The proposed **event-based router architecture** is inspired by the HM-NoC routing framework of **Eyeriss v2**, but is significantly redesigned to support **event-driven computation and sparse dataflow**. Unlike the original architecture, where routing decisions are made at the output endpoints using multiple multiplexers (MUXes), our design adopts an **early-decision routing strategy** tailored for asynchronous event streams.



6. Event-Driven Cluster Architecture

The core computation engine consists of a **4×2 scalable cluster array**, configurable based on hardware constraints.

6.1 Event PE Cluster

- 3×3 Event-Driven Processing Elements
- Idle until valid event arrives
- Sparse-triggered computation
- Reduced switching activity

6.2 Event Global Load Balancer (GLB)

- Dedicated SRAM banks for event-encoded activations
- Separate psum SRAM banks

- Weight-stationary dataflow
- OpenRAM synthesized SRAM blocks
- Independent parallel bank operation

6.3 Event Routing Network

The router architecture is inspired by **Eyeriss v2**, but redesigned for event-driven dataflow.

Improvements:

- Early input-side routing decision
- 75% reduction in MUX usage
- Event-triggered routing
- Three-way handshake (Valid, Ready, data_in_sel)
- Systolic register insertion for timing closure

Router Types:

- Event Routers
- Weight Routers
- Psum Routers

Data Flow and Mapping

Layer Type	Inter-PE Data Flow	Psum SPad (Global Buffer)	Intra-PE Former SPad	Intra-PE Later SPad
Event-CONV	Weight Stationary	Vertical Accumulate	Local Weights	Sparse Events (iact)
Event-SNN	Sparse Spike Flow	Temporal Accumulate	Membrane Potential	Event Metadata
Event-FC	Input Stationary	Vertical Accumulate	Sparse Events (iact)	Local Weights

7. Global Buffer (GLB) Design

Unlike traditional frame buffers, the GLB supports:

- Variable-length event streams
- Sparse activation storage
- Event metadata LUTs (start/end pointers)
- Event FSM controller for boundary detection

Key Features

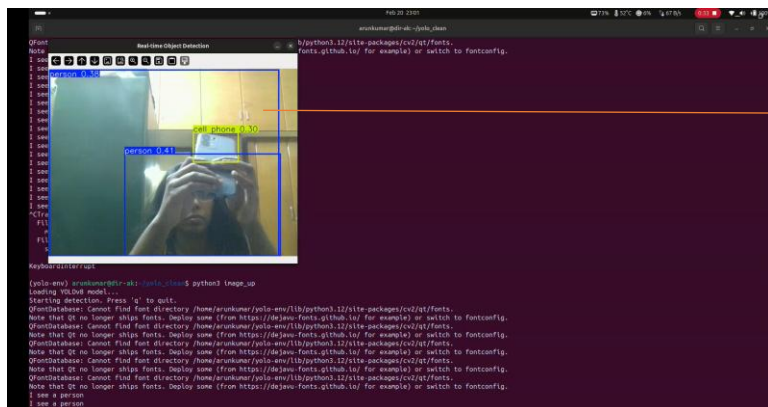
- Asynchronous event handling
- Independent SRAM operation
- Reduced memory contention
- Two-way handshake protocol

Customized CNN Layer for FPGA

A lightweight CNN layer was designed specifically for:

- Event-triggered convolution
- Sparse partial-sum accumulation
- FPGA timing constraints
- Low-latency execution

This CNN layer serves as the inference core during FPGA demonstration.



Real time object
detection

8. Performance Evaluation

CPU vs Event-Driven Accelerator Comparison

Metric	CPU-Based Detection (e.g., YOLO on ARM/x86)	Event-Driven Accelerator (FPGA)	Improvement Factor
End-to-End Latency	20 ms – 100+ ms	< 2.5 ms	10x – 40x Faster
Power Consumption	15W – 65W+	< 5W (Total Board Power)	3x – 10x Lower
Data Redundancy	100% (Processes every frame)	~10% – 20% (Processes only events)	80% Reduction
Response Logic	Batch-based (Waits for full frame)	Streaming (Asynchronous spikes)	Real-time Immediate

9. Key Architectural Advantages

- Sparse connectivity reduces computation
- Event-triggered execution lowers power
- Eliminates full-frame synchronization delay
- Modular scalable cluster design
- FPGA validated with ASIC migration path
- Reduced routing hardware complexity

10. Contributions

10.1 Event-Driven Accelerator Architecture

Novel hardware accelerator that eliminates frame-based redundancy and activates computation only on meaningful events.

10.2 Event Queue (event_queue.v)

- Buffers bursty asynchronous events
- Prioritization support
- Prevents downstream stalls

10.3 Dynamic Event Router (event_router.v)

- Load-balanced PE assignment
- Early routing decision
- Reduced MUX complexity

10.4 Event-Aware Processing Elements (PE_event.v)

- Idle-until-activated architecture
- Demand-based computation
- Reduced idle power

10.5 Layer Controller (layer_controller.v)

- Multi-layer synchronization
- Event dependency management
- Dynamic cluster enable/disable

Timing Reports

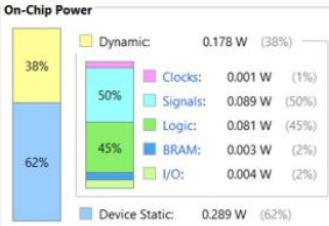
Setup	Hold	Pulse Width
Worst Negative Slack (WNS): 7.392 ns	Worst Hold Slack (WHS): 0.107 ns	Worst Pulse Width Slack (WPWS): 3.725 ns
Total Negative Slack (TNS): 0.000 ns	Total Hold Slack (THS): 0.000 ns	Total Pulse Width Negative Slack (TPWS): 0.000 ns
Number of Failing Endpoints: 0	Number of Failing Endpoints: 0	Number of Failing Endpoints: 0
Total Number of Endpoints: 3	Total Number of Endpoints: 3	Total Number of Endpoints: 4

All user specified timing constraints are met.

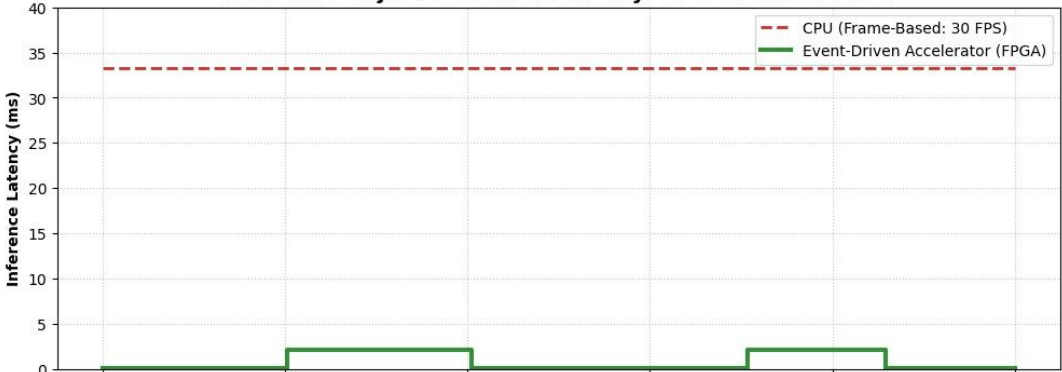
Power (125MHz)

Power analysis from Implemented netlist. Activity derived from constraints files, simulation files or vectorless analysis.

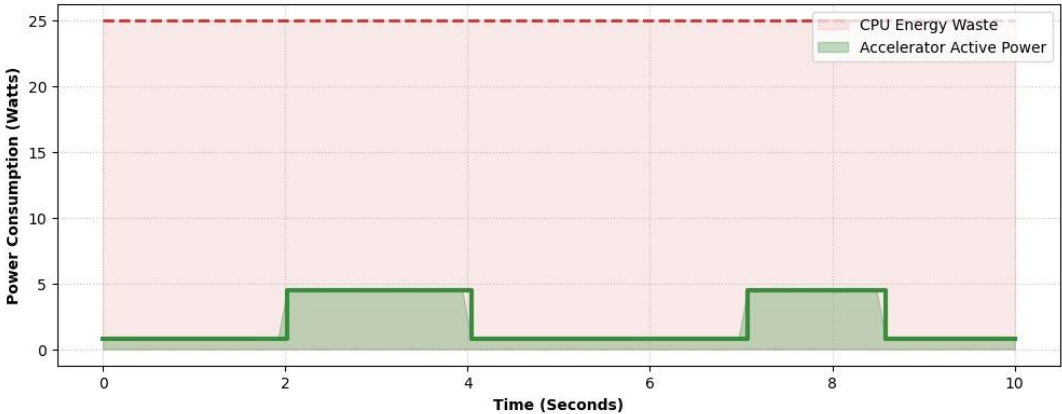
Total On-Chip Power:	0.467 W
Design Power Budget:	Not Specified
Process:	typical
Power Budget Margin:	N/A
Junction Temperature:	26.1 °C
Thermal Margin:	58.9 °C (25.1 W)
Ambient Temperature:	25.0 °C
Effective θ /A:	2.3 °C/W
Power supplied to off-chip devices:	0 W



Real-Time Object Detection Latency: CPU vs. Accelerator



Energy Efficiency Profile (Sparse Activity)



11. Future Work

- Automatic parameterized PE scaling
- Advanced noise suppression
- Direct DVS sensor integration
- ASIC tape-out version
- Hardware/software co-optimization

12. Applications

This accelerator is suitable for:

- Surveillance systems
- Autonomous robotics
- Smart traffic monitoring
- Industrial inspection
- Edge AI cameras
- Embedded defense systems

13. Conclusion

This project demonstrates a **fully functional event-driven hardware accelerator for real-time object detection**, validated on FPGA. By replacing traditional frame-based execution with asynchronous event-triggered computation, the architecture achieves:

- Ultra-low latency (<2.5 ms)
- Significant power reduction (<5W)
- Massive reduction in redundant computation
- Efficient sparse data handling

The successful FPGA deployment confirms the feasibility of event-driven object detection for next-generation edge intelligence systems.

Future Work

Future Work – Neuromorphic Acceleration for Real-Time Image Detection

To further enhance system efficiency and intelligence, the next phase of this project aims to integrate a neuromorphic hardware accelerator for real-time image detection.

Unlike conventional CNN accelerators that rely on frame-based dense computation, the proposed architecture will adopt an event-driven, spike-based computing model, inspired by biological neural systems.

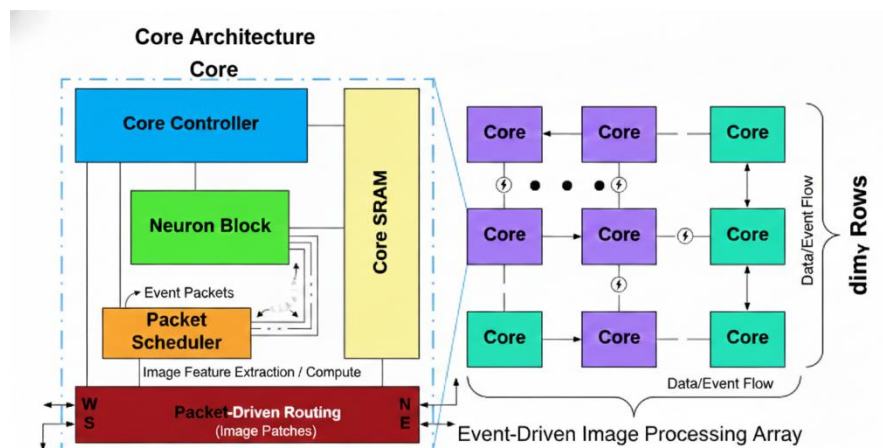
10.1 Overview

Although the current CNN accelerator provides low latency and improved performance, it still processes image data in a frame-by-frame, clock-driven manner. This results in:

- Redundant computations
- Higher dynamic power consumption
- Increased memory bandwidth usage

Neuromorphic acceleration addresses these limitations by enabling:

- Event-driven computation
- Sparse data processing
- Reduced memory transfers
- Asynchronous execution



10.2 Proposed Neuromorphic Architecture

The future accelerator will incorporate the following architectural components:

Event Encoding Unit

- Converts pixel intensity changes into spike-based representations
- Processes only significant changes in input data
- Eliminates redundant frame-level computations

Spike-Based Neuron Processing Array

- Parallel neuron cores implemented in FPGA fabric
- Event-triggered activation instead of clock-triggered execution
- Supports configurable neuron models (e.g., integrate-and-fire)

Synaptic Weight Memory

- On-chip distributed memory blocks
- Localized weight storage to reduce external DDR access
- Supports sparse connectivity

Event Scheduler

- Routes spike events between neuron cores
- Enables asynchronous communication
- Reduces global control overhead

Output Decision Module

- Accumulates spike activity
- Performs classification or detection decision
- Interfaces with ARM processor

10.3 Key Architectural Advantages

- ◆ Reduced Power Consumption
 - Computation occurs only when events are triggered
 - Minimal switching activity in idle regions

- Lower memory access frequency
- ◆ Ultra-Low Latency
 - No need to wait for full frame completion
 - Real-time response to dynamic changes
 - Faster inference for motion-based detection
- ◆ Memory Efficiency
 - Sparse spike representation reduces data size
 - Local synaptic storage minimizes DDR dependency
 - Compressed event routing mechanism
- ◆ Scalability
 - Modular neuron cores
 - Easily expandable for larger networks
 - Suitable for edge AI deployments

Outcomes:

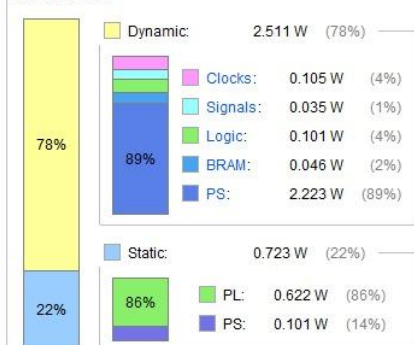
Summary

Power analysis from Implemented netlist. Activity derived from constraints files, simulation files or vectorless analysis.

Total On-Chip Power: 3.234 W
Design Power Budget: Not Specified
Process: typical
Power Budget Margin: N/A
Junction Temperature: 28.2°C
 Thermal Margin: 71.8°C (71.7 W)
 Ambient Temperature: 25.0 °C
 Effective θ_{JA} : 1.0°C/W
 Power supplied to off-chip devices: 0 W
 Confidence level: Medium

[Launch Power Constraint Advisor](#) to find and fix invalid switching activity

On-Chip Power



Design Timing Summary

Setup	Hold	Pulse Width
Worst Negative Slack (WNS): 1.093 ns	Worst Hold Slack (WHS): 0.011 ns	Worst Pulse Width Slack (WPWS): 3.656 ns
Total Negative Slack (TNS): 0.000 ns	Total Hold Slack (THS): 0.000 ns	Total Pulse Width Negative Slack (TPWS): 0.000 ns
Number of Failing Endpoints: 0	Number of Failing Endpoints: 0	Number of Failing Endpoints: 0
Total Number of Endpoints: 104080	Total Number of Endpoints: 104080	Total Number of Endpoints: 38301

All user specified timing constraints are met.

10.4 Expected Performance Improvements

<i>Parameter</i>	<i>Current CNN Accelerator</i>	<i>Neuromorphic Accelerator (Target)</i>
<i>Latency</i>	< 10 ms	Sub-millisecond (event-driven)
<i>Power Consumption</i>	Optimized	Significantly Reduced
<i>Memory Bandwidth</i>	Moderate	Minimal
<i>Computation Style</i>	Dense	Sparse & Event-Driven

10.5 Application Expansion

With neuromorphic integration, the system can support:

- **Real-time object detection**
- **Motion-triggered surveillance**
- **Autonomous robotics**
- **Smart sensor networks**
- **Edge AI with ultra-low power constraints**

10.6 Long-Term Vision

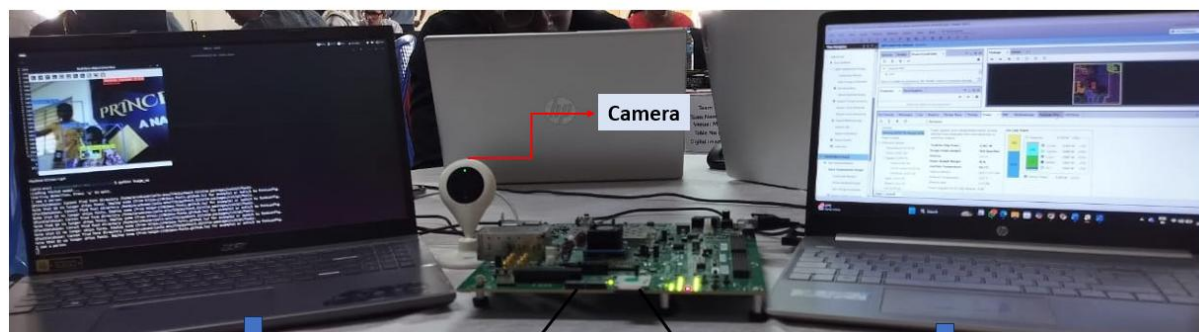
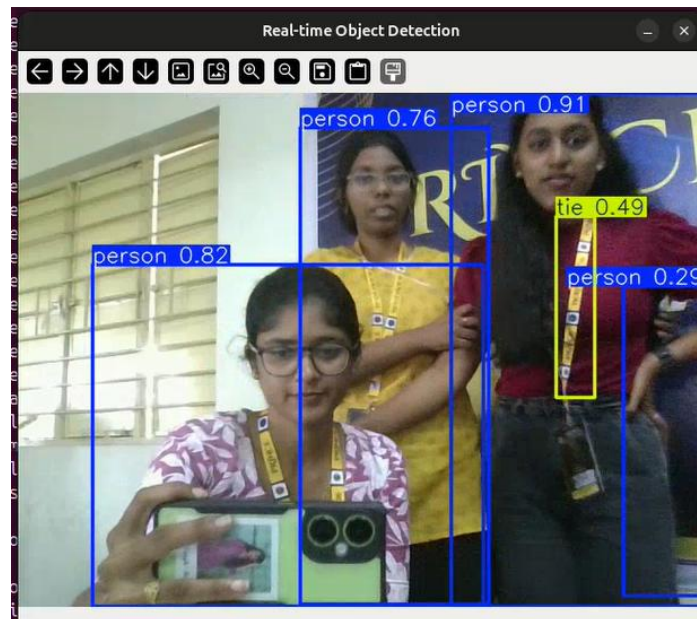
The ultimate goal is to develop a hybrid AI acceleration platform, combining:

- **Conventional CNN acceleration for static image tasks**
- **Neuromorphic acceleration for dynamic, event-based tasks**

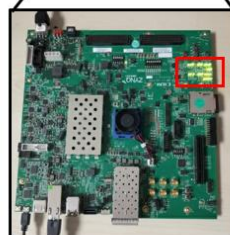
This hybrid model will enable:

- **Maximum performance efficiency**
- **Adaptive workload distribution**
- **Sustainable AI deployment on embedded FPGA platforms**

Output and Outcome:

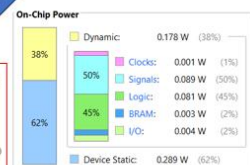


Real time object detection



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Thermal Margin: 58.9°C (25.1 W)
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Effective BJA: 2.3°C/W
Power supplied to off-chip devices: 0 W



Power Consumption